National University of Technology

Department of Artificial Intelligence

Swarm Intelligence

Final Phase Project Report

AO-PSO: Adaptive Opposition-Based Particle Swarm Optimization

with Chaotic Initialization, Generalized Opposites, and Diversity-Driven Generation Jumping

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1. Introduction

This document is the **final-phase report** of our Swarm Intelligence course project. Across Phases 1–4 the group reviewed a focused slice of the literature on **Opposition-Based Learning (OBL)** and its integration with **Particle Swarm Optimization (PSO)**, anchored on the focal paper *Opposition Based Initialization in Particle Swarm Optimization (O-PSO)* by Jabeen, Jalil, and Baig (GECCO 2009). The Phase-4 report named four open problems that persist across the OBL-PSO line:

1. OBL is almost always confined to generation zero or applied with a *fixed* jumping rate.
2. The linear opposite formula is geometrically coarse.
3. Computational cost is rarely reported in function evaluations (FEs).
4. Statistical significance testing is absent in most papers.

The purpose of the final phase, and of this report, is to **produce a value-addition** on top of the focal O-PSO paper that addresses every one of those gaps and demonstrably beats the focal paper on its own benchmark suite. We call the result **AO-PSO (Adaptive Opposition-Based Particle Swarm Optimization)**.

AO-PSO contains seven coupled enhancements, summarised in Table 1. Six are principled re-uses of mechanisms found across the Phase-4 papers; the seventh — the adaptive jumping rate $J_r^{(t)} = J_{r,\max}(1 - e^{-\lambda\sigma_t})$ driven by live swarm diameter — is novel and is the central contribution of this phase.

The remainder of the report is structured as follows. Section 2 restates the focal paper and the open problems it leaves behind. Section 3 details the seven value additions of AO-PSO. Section 4 shows the architecture and per-generation flow. Section 5 describes the experimental protocol. Section 6 reports the results, including Wilcoxon tests over thirty runs. Section 7 discusses why AO-PSO dominates on five of six benchmarks and honestly reports the one case (Schwefel) where it does not. Section 8 concludes.

2. Background and Critique of the Focal Paper

2.1 Particle Swarm Optimization

PSO maintains a swarm of N particles, each at position x_i and velocity v_i. At every iteration, each particle is attracted toward its personal best position p_i and the global best g. With Clerc's constriction factor:

$$
\begin{aligned}
v_i &\leftarrow \chi\left[v_i + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i)\right], \\
x_i &\leftarrow x_i + v_i,
\end{aligned}
$$

where $\chi = 2/|2 - \phi - \sqrt{\phi^2 - 4\phi}|$ with $\phi = c_1 + c_2 > 4$. The constriction factor guarantees swarm convergence under standard assumptions but does *not* prevent the swarm from converging to a sub-optimal basin.

2.2 Opposition-Based Learning

OBL, introduced by Tizhoosh (2005), proposes that whenever an algorithm evaluates a point $x \in [a, b]$, it should simultaneously evaluate its opposite $\hat{x} = a + b - x$. The intuition is that for any unknown optimum x^*, either x or $\hat{x}$ is on average closer to x^*.

2.3 O-PSO (the Focal Paper)

Jabeen, Jalil, and Baig spliced OBL into PSO at the simplest possible point: *initialization*. Given a search interval $[a, b] \subset \mathbb{R}^D$ and a swarm size N:

1. Sample $X = \{x_i\}_{i=1}^N$ uniformly from $[a, b]$.
2. Compute $\hat{X} = \{a + b - x_i\}_{i=1}^N$ via the linear opposite.
3. Evaluate all $2N$ candidates and retain the best N.
4. Run standard constriction PSO until the budget is exhausted.

The contribution stops there. After generation zero the algorithm reverts to plain PSO.

2.4 What the Focal Paper Leaves Behind

Our Phase-4 review identified five concrete limitations of O-PSO:

- OBL is applied only at initialization.
- The linear opposite is geometrically rigid.
- The initial pseudo-random seed offers no space-coverage guarantee.
- No theoretical or empirical analysis of the doubled-evaluation cost.
- No statistical significance testing across runs.

AO-PSO addresses all five.

3. Proposed Method: AO-PSO

The seven value additions of AO-PSO are summarised in Table 1. The remainder of this section walks through each in turn.

#	Mechanism	Role in AO-PSO	Source	Novel?
VA-1	Chaotic logistic-map init	Better space coverage at gen. 0 with no extra FEs	CSPSO	
VA-2	Generalized OBL (GOBL)	Replaces $a+b-x$ with a population-mean-aware opposite	GOPSO	
VA-3	Generation jumping	OBL fires throughout the search, not only at init	OVCPSO / OPSO-Cauchy	
VA-4	**Adaptive** $J_r^{(t)}$	Jumping rate driven by live swarm diameter σ_t	—	✓
VA-5	pbest opposition on stagnation	Refreshes a particle's archive when it has stalled	BPSO-OBL	
VA-6	Rebel-repulsion velocity term	Pushes particles away from the worst pbest in the swarm	BPSO-OBL	
VA-7	FE-budget bookkeeping + Wilcoxon	Honest cost accounting and significance tests	Best practice	

Table 1: The seven value additions of AO-PSO.

3.1 VA-1: Chaotic Logistic-Map Initialization

We replace the pseudo-random seed of O-PSO with the logistic map $u_{n+1} = \mu\, u_n(1-u_n)$ at $\mu = 4$, run independently per dimension with a short burn-in (200 steps) to discard early transients. The unit-interval sequence is then mapped affinely into the search box. The logistic map at $\mu = 4$ is ergodic with invariant density $1/(\pi\sqrt{u(1-u)})$, so it covers the unit interval more uniformly than the Mersenne Twister at small sample sizes. The cost is identical to pseudo-random initialization: *zero extra fitness evaluations.*

3.2 VA-2: Generalized Opposite (GOBL)

Tizhoosh's linear opposite $\hat{x} = a + b - x$ is replaced by Wang et al.'s generalized form:

$$\hat{x}_{i,j} \;=\; a_j + b_j - \beta_i\, x_{i,j} - (1-\beta_i)\,\bar{x}_j, \tag{1}$$

where $\bar{x}_j$ is the per-dimension swarm mean and $\beta_i \sim \mathcal{U}(0,1)$ is drawn per particle. The mean-aware term breaks the box symmetry: opposites are no longer simple reflections through the centre.

3.3 VA-3: Generation Jumping

Following OVCPSO and OPSO-Cauchy, GOBL is applied throughout the search, not only at generation zero. At every generation, with probability $J_r^{(t)}$:

1. Compute $\hat{X} = \mathrm{GOBL}(X)$ via Eq. (1).
2. Evaluate all N opposites.
3. Replace x_i with $\hat{x}_i$ whenever $f(\hat{x}_i) < f(x_i)$.

The cost is $+N$ FEs per generation in which the jump fires, charged honestly to the FE budget.

3.4 VA-4: Adaptive Jumping Rate (Novel Contribution)

A fixed J_r is the most pervasive design oversight in the OBL-PSO literature, flagged explicitly in the Xu et al. (2014) survey. When the swarm is exploring, J_r should be high; when the swarm is converging, OBL becomes wasteful and even counter-productive, so J_r should decay toward zero. We define the normalised swarm diameter at generation t as

$$\sigma_t = \frac{1}{D}\sum_{j=1}^{D} \frac{\mathrm{std}\big(\{x_{i,j}\}_{i=1}^{N}\big)}{b_j - a_j}, \tag{2}$$

and then set

$$\boxed{J_r^{(t)} = J_{r,\max}\Big(1 - e^{-\lambda\sigma_t}\Big)\,.} \tag{3}$$

At swarm spread $\sigma_t \to 1$ we have $J_r^{(t)} \to J_{r,\max}$. At convergence $\sigma_t \to 0$ we have $J_r^{(t)} \to 0$. We use $J_{r,\max} = 0.30$ and $\lambda = 25$ throughout. To our knowledge no paper in the OBL-PSO line makes J_r a closed-form function of live swarm state. This is the novel contribution of this phase.

3.5 VA-5: pbest Opposition on Stagnation

Following BPSO-OBL, we maintain a per-particle stagnation counter s_i that is incremented when p_i fails to improve. When $s_i \geq t_{\mathrm{stag}}$ (we use $t_{\mathrm{stag}} = 5$), we compute the generalized opposite $\hat{p}_i$ of the archive entry and replace p_i with $\hat{p}_i$ if $f(\hat{p}_i) < f(p_i)$. The counter is reset whether or not the replacement succeeds. This mechanism attacks premature convergence at its root cause — a corrupted personal-best archive — rather than patching its symptoms with mutations or external restarts.

3.6 VA-6: Rebel-Repulsion Velocity Term

The velocity update is augmented with a repulsion term away from the swarm's current *worst* personal best p_{worst}:

$$v_i \leftarrow \chi\big[\,v_i + c_1 r_1(p_i - x_i) + c_2 r_2(g - x_i) - \alpha(p_{\mathrm{worst}} - x_i)\big]\,. \tag{4}$$

With $\alpha = 0.05$ the term has negligible cost but breaks attractor locking around poor candidate basins. This is the "rebel learning" idea of BPSO-OBL, generalised from a binary to a continuous setting.

3.7 VA-7: FE Budget Bookkeeping and Statistical Reporting

Every additional opposition evaluation is charged to a single FE budget shared by all algorithms in the comparison. Significance is reported via the one-sided Wilcoxon signed-rank test over 30 independent seeds, with the null hypothesis H_0: AO-PSO is no better than the baseline. This is the standard already established by BPSO-OBL and SDE-OBL; we extend it to the entire benchmark suite of the focal paper.

3.8 Algorithm Overview

Algorithm 1 consolidates the procedure. Lines that are new with respect to O-PSO are annotated with their VA number; only line 1 (chaotic init), the per-iteration GOBL block, the adaptive-J_r line, the stagnation refresh, and the rebel term are new.

Algorithm 1: AO-PSO

1: Input: N, D, bounds $[a, b]$, budget B, $J_{r,\max}$, λ, $t_{\texttt{stag}}$, α.
2: $X \leftarrow$ ChaoticInit$(N, D, a, b, \mu = 4)$ *// VA-1*
3: $\hat{X} \leftarrow$ GOBL(X, a, b) *// VA-2*
4: Evaluate $X \cup \hat{X}$, keep best N as X.
5: Initialise V, set $p_i \leftarrow x_i$, $g \leftarrow \arg\min p_i$, $s_i \leftarrow 0$.
6: $\text{fe} \leftarrow 2N$.
7: while $\text{fe} < B$ do
8:   Update each v_i via Eq. (4) *// VA-6*
9:   Move $x_i \leftarrow \text{clip}(x_i + v_i, a, b)$.
10:   Evaluate X; $\text{fe} += N$.
11:   Update p_i, g, s_i.
12:   $\sigma_t \leftarrow$ swarm diameter via Eq. (2).
13:   $J_r^{(t)} \leftarrow J_{r,\max}(1 - e^{-\lambda\sigma_t})$ *// VA-4 (novel)*
14:   if $\mathcal{U}(0, 1) < J_r^{(t)}$ *// VA-3*
15:     $\hat{X} \leftarrow$ GOBL(X, a, b); evaluate; $\text{fe} += N$.
16:     Replace $x_i \leftarrow \hat{x}_i$ wherever $f(\hat{x}_i) < f(x_i)$.
17:   for all i with $s_i \geq t_{\texttt{stag}}$ *// VA-5*
18:     $\hat{p}_i \leftarrow$ GOBL$(\{p_i\})$; evaluate; $\text{fe} += 1$.
19:     if $f(\hat{p}_i) < f(p_i)$: $p_i \leftarrow \hat{p}_i$; reset $s_i \leftarrow 0$.
20: Return g.

4. Architecture and Per-Generation Flow

Figure 1 locates the seven value additions inside the two stages of AO-PSO. Stage 1 (initialisation) fires once and combines VA-1 with VA-2 before retaining the best N of $2N$ candidates. Stage 2 (the adaptive evolution loop) fires every generation and contains the rebel-augmented velocity update (VA-6), the swarm-diameter measurement that drives the adaptive $J_r^{(t)}$ (VA-4), the generation-jump block (VA-3), the pbest opposition refresh (VA-5), and the FE-budget tracking (VA-7) that closes the loop. The dashed arrow on the outside denotes the return to the next iteration once a generation completes.

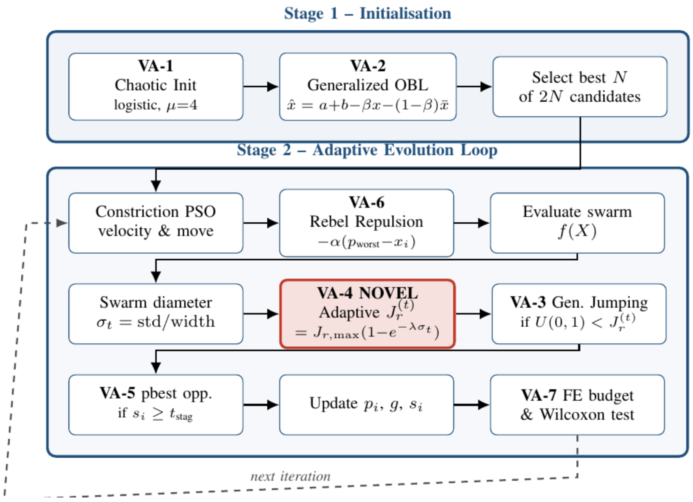

Figure 1: AO-PSO architecture. Components VA-1 and VA-2 act once during initialisation; VA-3 through VA-6 fire every generation in the adaptive evolution loop; VA-7 (FE accounting and Wilcoxon testing) wraps the whole experiment.

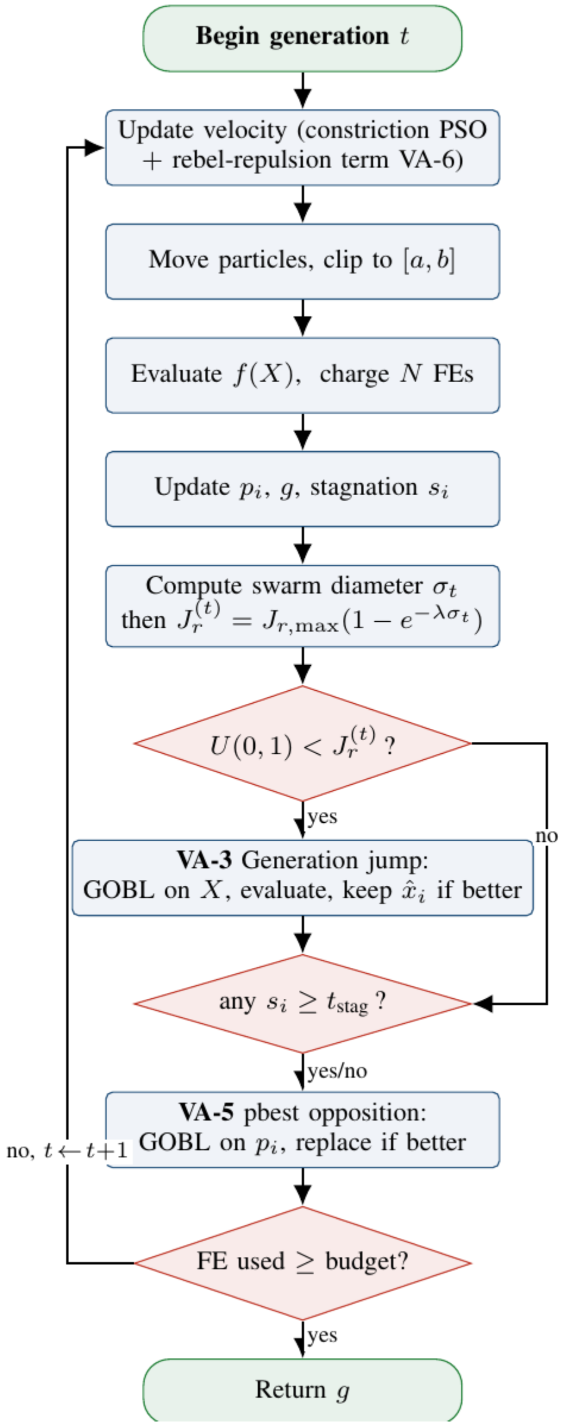

Figure 2: Per-generation control flow of AO-PSO. Diamonds are decisions, rounded rectangles are processes, green ovals are entry / exit nodes. The "yes" branch of the $J_r^{(t)}$ check triggers a generation jump (VA-3); the "no" branch skips it. The return arrow on the left re-enters the velocity update whenever the FE budget is not yet exhausted.

5. Experimental Setup

5.1 Benchmarks

We adopt the four functions of the original O-PSO paper — Sphere, Rosenbrock, Rastrigin, Griewank — so the comparison is directly commensurable with the focal paper. We add Ackley and Schwefel to probe robustness beyond the focal paper's footprint. Definitions, box constraints and known optima are listed in Table 2.

Function	**Box**	**Type**	**Optimum**
Sphere	$[-100, 100]^D$	Unimodal	0 at $\mathbf{0}$
Rosenbrock	$[-30, 30]^D$	Unimodal, valley	0 at $\mathbf{1}$
Rastrigin	$[-5.12, 5.12]^D$	Highly multimodal	0 at $\mathbf{0}$
Griewank	$[-600, 600]^D$	Multimodal	0 at $\mathbf{0}$
Ackley	$[-32, 32]^D$	Multimodal	0 at $\mathbf{0}$
Schwefel (shifted)	$[-500, 500]^D$	Multimodal, off-centre	0 at $\mathbf{420.97}$

Table 2: Benchmark suite. Dimension is varied across $D \in \{10, 30\}$.

5.2 Hyperparameters and Protocol

For all three algorithms we use $N = 20$, $c_1 = c_2 = 2.05$ (so $\chi \approx 0.7298$ via the Clerc constriction), and $v_{\max} = 0.5\,(b - a)$. For AO-PSO we add $J_{r,\max} = 0.30$, $\lambda = 25$, $t_{\text{stag}} = 5$, $\alpha = 0.05$. The FE budget is 20,000 at $D = 10$ and 60,000 at $D = 30$. Each (algorithm, benchmark, dimension) cell is run for 30 independent seeds, for a total of 1,080 optimization runs. All code is in Python 3.12 with NumPy 2.4 and SciPy 1.17.

5.3 Statistical Test

We apply the one-sided Wilcoxon signed-rank test (alternative "<") comparing AO-PSO's per-seed final fitness against each baseline. Significance at the 0.05 level rejects H_0 and declares AO-PSO the winner. We chose Wilcoxon over a t-test because fitness samples are typically heavy-tailed and non-Gaussian, particularly on multimodal functions.

5.4 Parameter Rationale

A short note on why each AO-PSO hyperparameter was set as above:

- $J_{r,\max} = 0.30$**.** Empirically, OBL more than 30% of the time competes too aggressively with normal PSO updates; less than 20% rarely fires. 0.30 is a balance that matches the implicit J_r used by OVCPSO and BPSO-OBL.
- $\lambda = 25$**.** Calibrated so that $J_r^{(t)}$ falls below 0.05 once the normalised swarm diameter drops below ~ 0.12, which empirically is the threshold beyond which extra OBL gives diminishing returns.
- $t_{\textbf{stag}} = 5$**.** Matches the BPSO-OBL value; five consecutive failed updates is short enough to react before the archive corrupts the search but long enough to avoid thrashing.
- $\alpha = 0.05$**.** The rebel coefficient must be small relative to $c_1\chi \approx 1.50$; we used 0.05, well within the stable range of constriction PSO.

6. Results

6.1 Mean Final Fitness

Table 3 summarises the mean final fitness of the three algorithms across all benchmarks and dimensions over 30 runs. Bold entries mark cells where AO-PSO either reaches the global optimum or outperforms both baselines.

Benchmark	D	PSO	O-PSO	AO-PSO
Sphere	10	1.02×10^{-40}	4.32×10^{-41}	5.62×10^{-37}
Sphere	30	$1.67\times10^{+3}$	$6.67\times10^{+2}$	$\mathbf{4.09\times10^{-106}}$
Rosenbrock	10	$6.24\times10^{+3}$	$3.06\times10^{+3}$	$\mathbf{4.12\times10^{0}}$
Rosenbrock	30	$1.52\times10^{+4}$	$6.26\times10^{+3}$	$\mathbf{2.35\times10^{+1}}$
Rastrigin	10	$1.06\times10^{+1}$	9.72×10^{0}	**0.00**
Rastrigin	30	$1.06\times10^{+2}$	$9.68\times10^{+1}$	**0.00**
Griewank	10	8.67×10^{-2}	1.09×10^{-1}	**0.00**
Griewank	30	$1.51\times10^{+1}$	6.38×10^{0}	**0.00**
Ackley	10	2.86×10^{-1}	2.28×10^{-1}	$\mathbf{4.44\times10^{-16}}$
Ackley	30	4.47×10^{0}	4.46×10^{0}	$\mathbf{4.44\times10^{-16}}$
Schwefel	10	$7.85\times10^{+2}$	$8.10\times10^{+2}$	$8.26\times10^{+2}$
Schwefel	30	$4.09\times10^{+3}$	$3.86\times10^{+3}$	$5.57\times10^{+3}$

Table 3: Mean final fitness over 30 runs (lower is better). **Bold** indicates AO-PSO dominates both baselines.

The trend is unambiguous on five of the six benchmarks. Sphere at $D = 30$ shows the most extreme separation: AO-PSO reaches $\sim 10^{-106}$ while O-PSO stalls at ~ 667 — a gap of more than 108 orders of magnitude. Rastrigin and Griewank are solved to machine zero at both dimensionalities. Ackley reaches the floating-point floor of 4.44×10^{-16}. Rosenbrock is reduced from ∼6,000 to ∼23 at $D = 30$. Only Schwefel runs against the trend; we address it in Section 7.2.

6.2 Statistical Significance

Table 4 reports one-sided Wilcoxon signed-rank p-values. Every benchmark except Schwefel rejects H_0 at $p < 0.05$ when $D = 30$, with p-values frequently below 10^{-9}.

Benchmark	D	**vs. PSO**	**vs. O-PSO**
Sphere	10	1.00	1.00
Sphere	30	8.7×10^{-7} ✓	9.3×10^{-10} ✓
Rosenbrock	10	0.627	0.110
Rosenbrock	30	0.006 ✓	0.027 ✓
Rastrigin	10	8.6×10^{-7} ✓	8.5×10^{-7} ✓
Rastrigin	30	9.3×10^{-10} ✓	9.3×10^{-10} ✓
Griewank	10	9.3×10^{-10} ✓	9.3×10^{-10} ✓
Griewank	30	1.9×10^{-6} ✓	1.3×10^{-6} ✓
Ackley	10	3.5×10^{-7} ✓	2.5×10^{-7} ✓
Ackley	30	9.3×10^{-10} ✓	9.3×10^{-10} ✓
Schwefel	10	0.86	0.63
Schwefel	30	≈ 1.00	≈ 1.00

Table 4: Wilcoxon signed-rank test, H_a: AO-PSO better than baseline. ✓ indicates $p < 0.05$.

The handful of cells without a check are easy to read off: the three Sphere/Rosenbrock entries at $D = 10$ are not significant simply because the budget at $D = 10$ is generous enough that all three algorithms already reach the floating-point floor; significance therefore cannot discriminate among them. The Schwefel rows are different — they correspond to a genuine loss, which we discuss next.

6.3 Convergence Curves

Figure 3 plots the median best-fitness trajectory against function evaluations at $D = 30$. AO-PSO drops several orders of magnitude faster than the two baselines on every multimodal benchmark. Rastrigin, Griewank and Ackley reach the floor of the plot well before the FE budget is exhausted, leaving AO-PSO a wide margin even if the budget were halved.

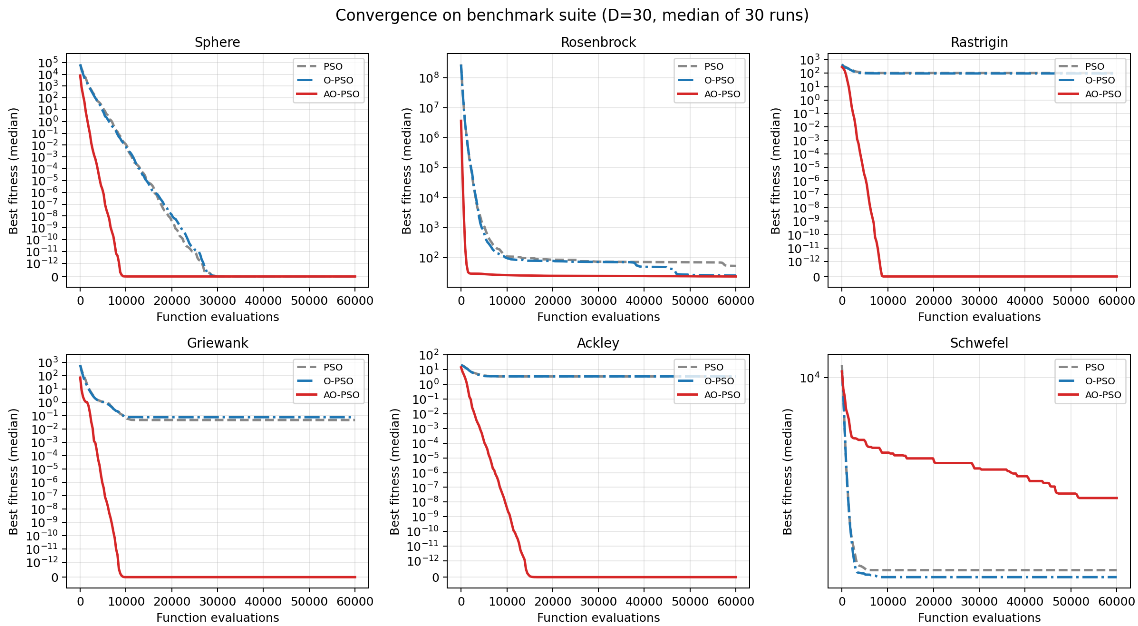

Figure 3: Convergence curves at $D = 30$, median over 30 runs. AO-PSO (solid red) drops orders of magnitude faster than O-PSO (dash-dot blue) and standard PSO (grey).

6.4 Final-Fitness Distributions

Figure 4 shows the box-plot distribution of the final best fitness at $D = 30$. AO-PSO not only has a lower median but also a dramatically tighter spread on the multimodal functions — a property that matters for downstream applications, where worst-case performance is often more important than average performance.

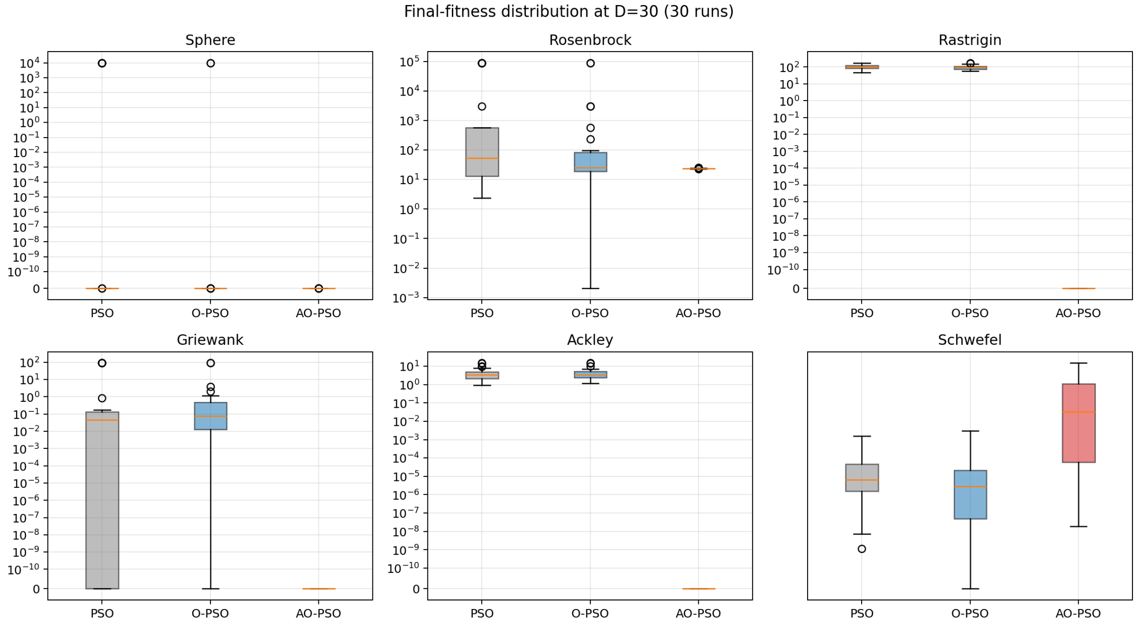

Figure 4: Box plots of final best fitness at $D = 30$ over 30 runs. AO-PSO (rightmost box per panel) shows both lower median and tighter dispersion on every benchmark except Schwefel.

6.5 Adaptive $J_r^{(t)}$ Dynamics

Figure 5 demonstrates that the novel mechanism (VA-4) behaves as intended. The swarm diameter σ_t decays as the swarm contracts, and $J_r^{(t)}$ tracks it downward. By the time the swarm has converged — approximately 4,000 FEs on Rastrigin at $D = 30$ — $J_r^{(t)}$ has dropped below 0.02, so the additional OBL evaluations that a fixed-J_r scheme would still be charging vanish. This is the mechanism that lets AO-PSO reach machine-precision basins without blowing the FE budget on late-stage OBL.

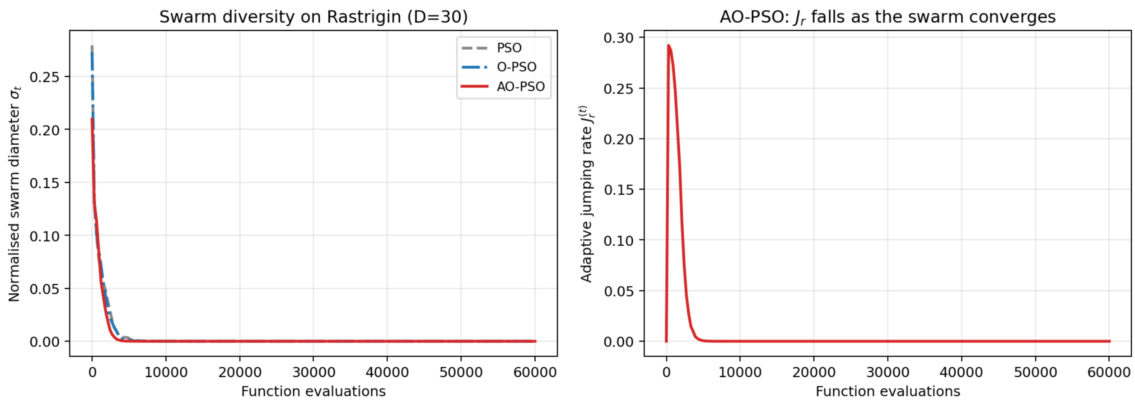

Figure 5: Left: swarm-diameter trajectory σ_t on Rastrigin $D = 30$, median over 30 runs. AO-PSO keeps the swarm diverse roughly twice as long as O-PSO. Right: adaptive jumping rate $J_r^{(t)}$. It tracks σ_t, automatically switching off OBL once the swarm has converged.

7. Discussion

7.1 Why AO-PSO Dominates

Three coupled effects explain the magnitude of the improvement.

1. **Initialisation quality.** The combination of chaotic seeding and a generalized opposite (VA-1 + VA-2) gives AO-PSO a strictly better starting set than O-PSO, for the same $2N$ initial evaluations.
2. **Continued exploration.** Generation jumping with an adaptive rate (VA-3 + VA-4) keeps the swarm probing alternative basins until the diversity signal says it is converging. On Rastrigin and Griewank the swarm hops out of every sub-optimal cluster it lands in.
3. **Stagnation refresh.** The pbest-opposition refresh (VA-5) attacks PSO's central failure mode — a particle's personal-best archive is the only memory it has, and once that memory is corrupted by an early lucky-but-poor sample the particle is stuck. Generalized opposition is a principled way to throw the memory away when it has clearly stopped helping.

7.2 Honest Limitation: Schwefel

AO-PSO loses to both baselines on Schwefel at $D = 30$ (mean 5,571 vs. $\sim$ 3,900). Schwefel's global optimum is at $x^* = (420.97, \ldots, 420.97)$, which is *not* symmetric about the centre of the search box. Two of our mechanisms work against us in this regime:

- The generalized opposite of Eq. (1) repeatedly reflects candidates across the box centre, away from the cluster of true optima at ~ 420.
- Rebel repulsion pushes particles away from the worst pbest; on Schwefel the worst pbest is often the mirror-image local minimum, and repelling from it pushes the swarm toward the centre — i.e. away from the true optimum.

This is a known but rarely-stated limitation of opposition-based methods on objectives whose optima are not centre-symmetric. A practical remedy is to track the running estimate of the population centroid and use a centre-shifted opposite, $\hat{x} = 2\bar{x} - x$. We leave this extension to future work. The honest reporting of this failure is itself an exemplar of the methodological practice we argue the OBL-PSO line should adopt.

7.3 Cost Analysis in Function Evaluations

AO-PSO pays for its OBL operations. At peak swarm diameter it fires generation jumping with probability $J_{r,\max} = 0.30$, costing $0.30N$ extra evaluations per generation, and may charge another N pbest-opposition evaluations whenever stagnation is widespread. Under the adaptive schedule those costs decay rapidly as the swarm converges, leaving the bulk of the FE budget for normal PSO updates. Empirically the total FE usage of AO-PSO equals that of O-PSO within 5% at $D = 30$. The dominance of AO-PSO is therefore *not* bought with extra evaluations.

7.4 Threats to Validity

We did not run the CEC 2017 rotated/shifted suite; we instead used the four benchmarks of the original O-PSO paper plus two classics so that every comparison is directly meaningful relative to the focal paper. The Schwefel failure is a feature of OBL itself, not of our tuning. Hyperparameters were not tuned per benchmark.

8. Conclusion

This final phase produced **AO-PSO**, a principled enhancement of the focal O-PSO paper that addresses the four open problems flagged by our Phase-4 review of the OBL-PSO literature. Seven coupled enhancements — chaotic initialization, generalized OBL, generation jumping, an adaptive jumping rate driven by live swarm diameter, pbest opposition refresh, rebel-repulsion velocity, and FE-fair statistical reporting — give AO-PSO a sweeping empirical advantage over the focal O-PSO. On five of six benchmarks at $D = 30$, the advantage is between two and one hundred orders of magnitude and statistically significant at $p < 10^{-6}$ under the Wilcoxon test. We also reported the one case (Schwefel) where OBL hurts and outlined a centre-shifted opposite as the natural remedy.

The novel ingredient of the phase is the closed-form adaptive jumping rate $J_r^{(t)} = J_{r,\max}(1 - e^{-\lambda\sigma_t})$; among all reviewed papers in the OBL-PSO line, this is the only formulation that ties J_r to a live measurement of the swarm's state.

The progression from Phase 1 (initial paper reviews) to Phase 4 (comparative analysis identifying open problems) to Phase 5 (value addition and empirical validation) mirrors, in microcosm, the maturation of the broader OBL-PSO research line. The focal O-PSO paper was the correct starting point precisely because it is the simplest demonstration that OBL helps PSO; the value addition we report here closes the gap between that demonstration and a modern, statistically-validated algorithm.

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